Computer Networks Worksheet

Executive Summary

The purpose of this project is to design, implement, and validate a **Layer 3 switched network** for MotorPH's physical headquarters. The network is structured to provide secure, efficient, and scalable connectivity for multiple departments—Sales, HR, Marketing, Accounting, and IT—while ensuring high availability and simplified management.

A **Core–Access network design** was adopted, with a Layer 3 Core Switch handling **inter-VLAN routing** and DHCP services. Access switches connect departmental endpoints via VLAN-segmented ports, with uplinks configured as 802.1Q trunks. The Core Switch acts as the gateway for all internal traffic between VLANs, while the Edge Router handles traffic leaving the network to external destinations.

During implementation, several common networking issues were identified, including DHCP failures, trunk encapsulation errors, and external connectivity problems. These were successfully resolved, and the network was verified through connectivity tests, DHCP assignment validation, and inter-VLAN ping tests.

The final network design demonstrates **high performance, scalability, and reliability**, meeting all business requirements for MotorPH. This report documents the design rationale, configuration details, troubleshooting strategies, and risk mitigation measures, providing a clear reference for both operational and future expansion needs

Part 1: Physical Layer Design

This section documents devices, media, and physical connectivity of the network.

Component	Detail	Rationale / Verification
Active Network Devices	Core Switch (L3), 5 Access Switches (L2), 1 Router (R1)	Verified device models and counts match the physical topology diagram.
Connecting Media	Copper Straight-Through Ethernet Cables (Primary)	Standard media for modern LAN connectivity.

Physical Topology	Extended Star Topology	Central Core Switch is the hub for all Access Switches, extending the network outward to endpoints.
Interface Types	GigabitEthernet is used for uplinks where available; in the lab, Core Switch → Access Switch links use FastEthernet due to limited GigabitEthernet ports. End-device ports on Access Switches are FastEthernet.	Uplinks carry aggregated traffic from multiple devices to the Core Switch, while end-device ports provide access for PCs and printers. This setup ensures full connectivity and reflects the actual Packet Tracer topology.

Part 2: Data Link Layer Design

This section documents VLANs, trunking, and Layer 2 protocols.

Component	Detail	Rationale / Verification
VLANs Defined	VLANs 100 (Sales), 103 (HR), 104 (Marketing), 106 (Accounting), 107 (IT)	Segregates traffic and controls broadcasts; essential for security and efficiency.
VLAN Tagging Protocol	IEEE 802.1Q	Standard protocol to carry multiple VLANs over a single trunk link.
Trunk Links	5 Trunk Links (Core-SW → Access Switches)	Allows all VLAN traffic to pass between Core and Access layer.
Spanning Tree Protocol	PVST (spanning-tree mode pvst)	Prevents Layer 2 loops in redundant paths.

Part 3: Network Layer Design – IP Addressing Scheme

A. Overall Network Address Summary

Type	Class	Address Range	Subnetting Plan
Private (RFC 1918)	Class C	192.168.10.0 – 192.168.10.255	VLSM used to create smaller, efficiently sized subnets per department.
Simulated Destination	Public-like (simulated)	8.8.8.8	/32 Loopback address for testing external connectivity.

B. Detailed Subnetting Plan (VLSM)

Department	VLAN ID	Subnet Address	Subnet Mask (CIDR)	SVI Gateway	Reserved / Unused VLAN Ports
Sales	100	192.168.10.0	/27 (255.255.255.224)	192.168.10.1	Fa0/17 – Fa0/24
HR	103	192.168.10.32	/28 (255.255.255.240)	192.168.10.33	Fa0/5 – Fa0/15
Marketing	104	192.168.10.48	/28 (255.255.255.240)	192.168.10.49	Fa0/7 – Fa0/15
Accounting	106	192.168.10.64	/29 (255.255.255.248)	192.168.10.65	Fa0/5 – Fa0/15
IT	107	192.168.10.72	/29 (255.255.255.248)	192.168.10.73	Fa0/3 – Fa0/10
Router Link (Internal)	N/A	192.168.1.0	/30 (255.255.255.252)	192.168.1.1 (Core-SW)	N/A

Note: SVI gateway addresses are assigned to the Core-SW, enabling inter-VLAN routing. External connectivity is handled via the Edge Router (R1).

Part 4: Edge & External Connectivity (Routing)

Component	Detail	Rationale / Verification
Routing Architecture	Layer 3 Switching (ip routing on Core-SW)	High-speed, hardware-based routing between internal VLANs.
Edge Device Role	R1 acts as the simulated Internet host	Uses Loopback8 (8.8.8.8) for external testing; no real NAT or internet required.
NAT Type	Not Required	Internal routing only; external testing uses simulation.
External Routing (Core-SW)	Static default route: ip route 0.0.0.0 0.0.0.0 192.168.1.2	Directs all external-bound traffic to R1.
Internal Routing (R1)	Static route back to LAN: ip route 192.168.10.0 255.255.255.0 192.168.1.1	Ensures replies from R1 reach the appropriate VLANs through Core-SW.

Part 5: Risk Assessment & Mitigation

Issue Encountered	Cause Identified	Resolution Strategy
Total DHCP Failure (APIPA addresses)	VLAN not created on Access Switch	Added VLAN manually using vlan [ID] command.
Trunking Command Rejection	Core-SW required explicit trunk encapsulation	Applied switchport trunk encapsulation dot1q before enabling trunk.
External Ping Failure (8.8.8.8)	Missing default route on Core-SW or missing return route on R1	Validated internal path (ping 192.168.1.2); ensured static route back to 192.168.10.0/24 on R1.

Network Diagram:

- https://drive.google.com/file/d/1TsR66PjWupQyqEBQVG4xkqE5g-HnLg0W/view?usp=sharing